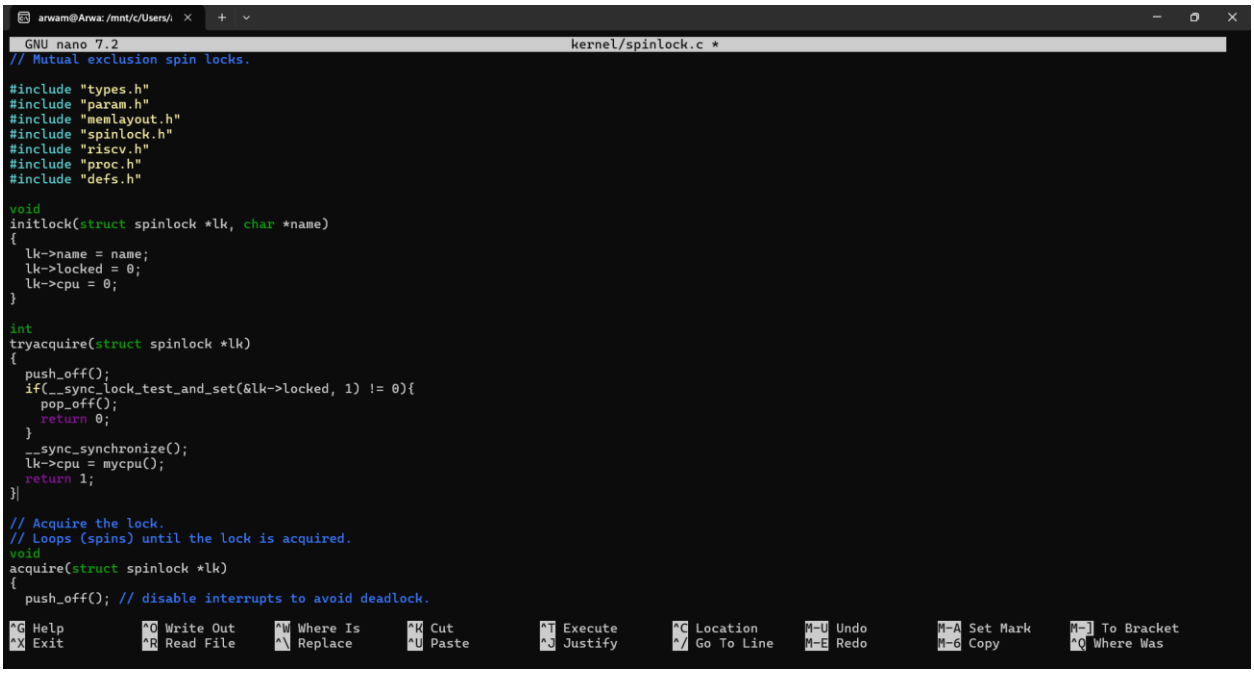
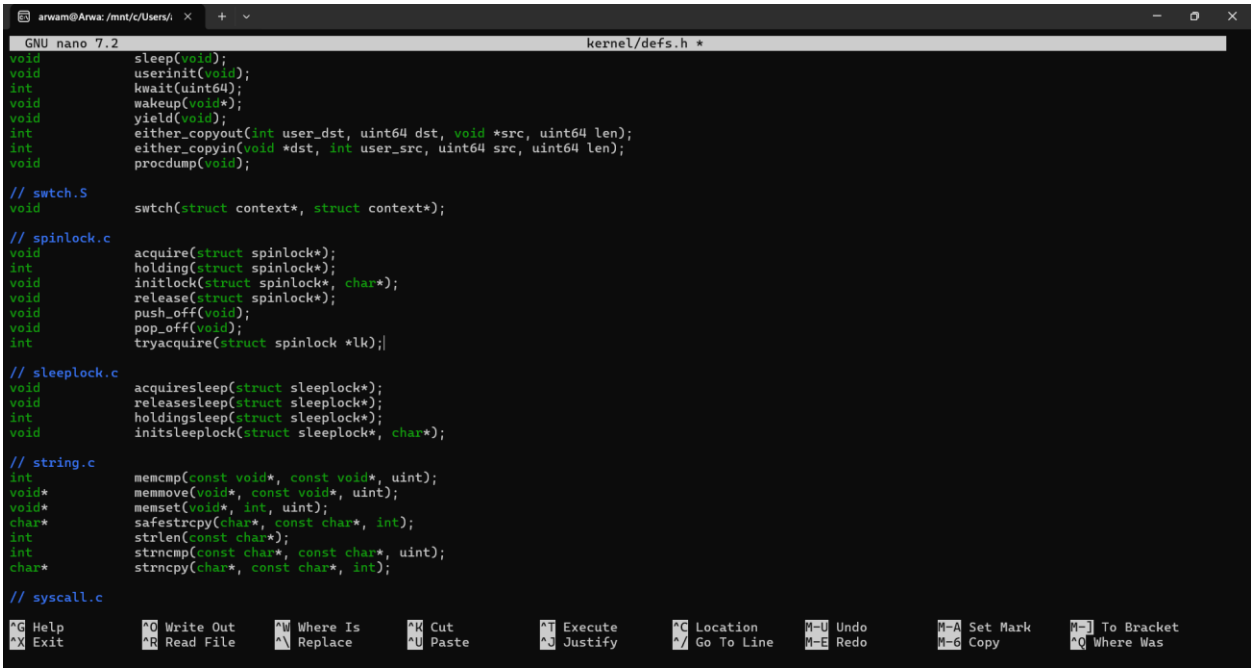
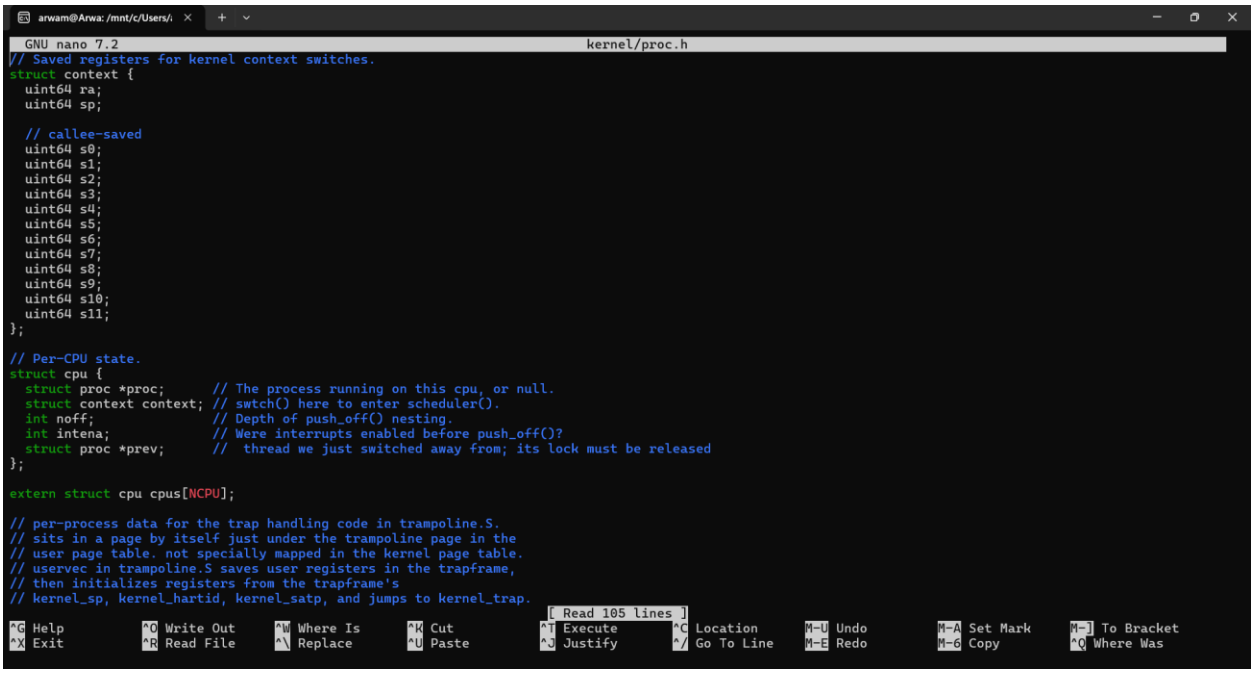
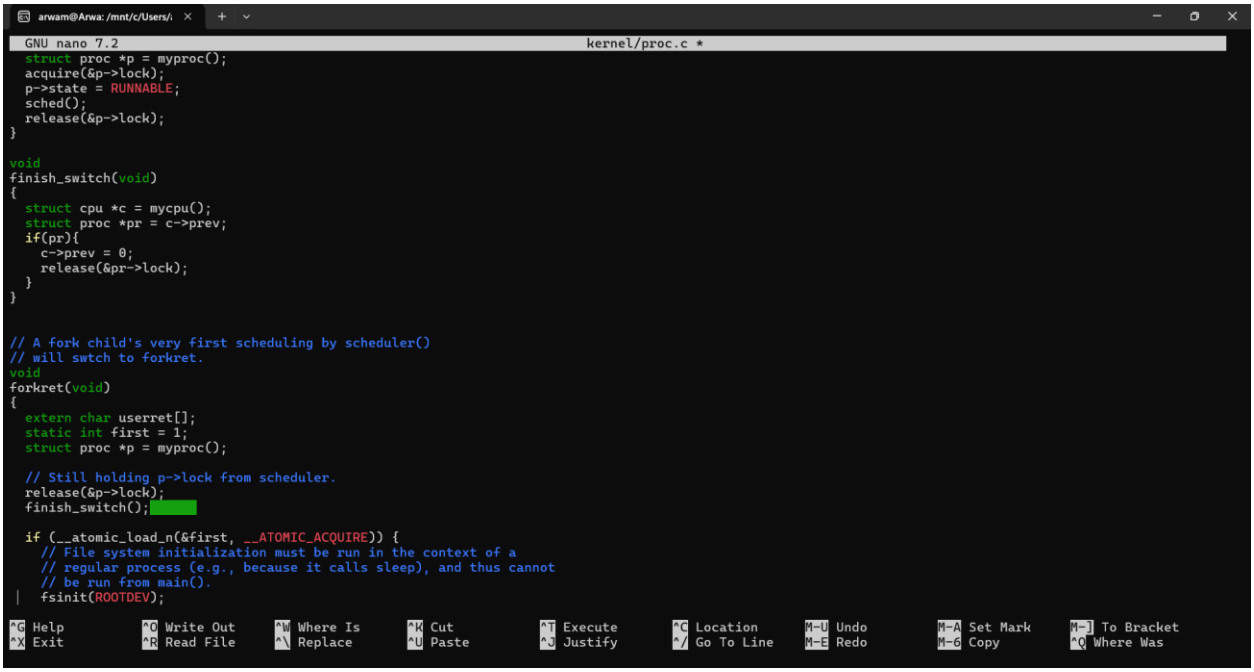
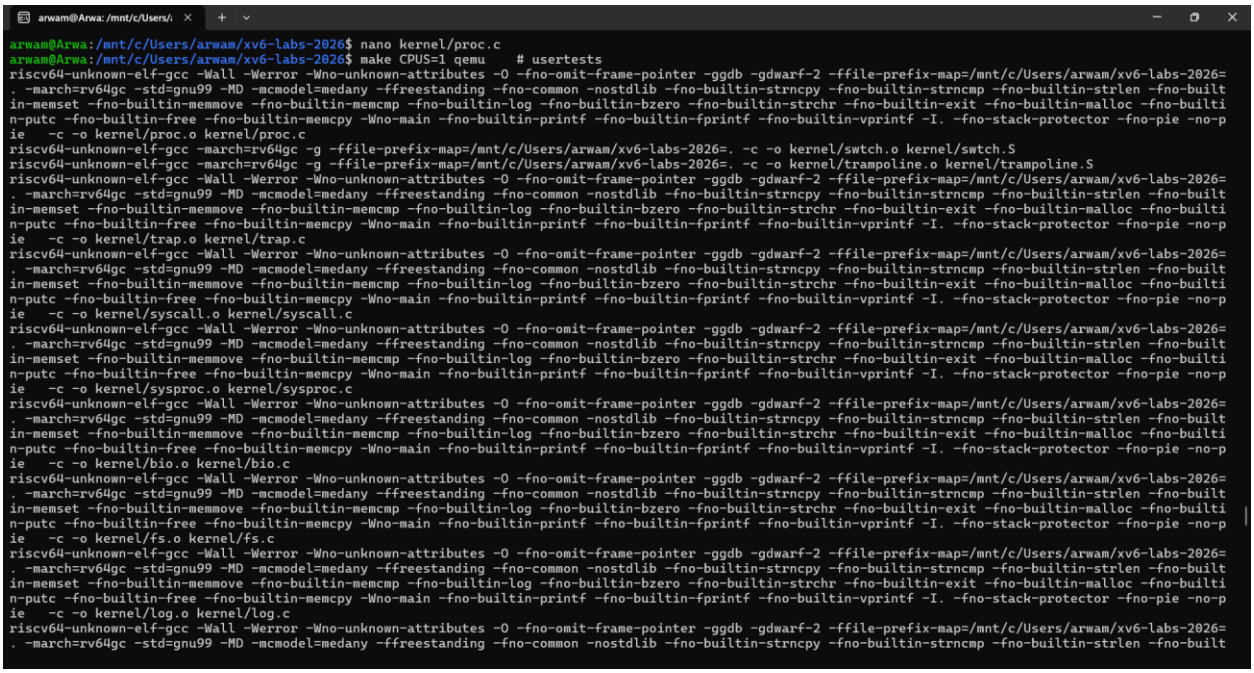
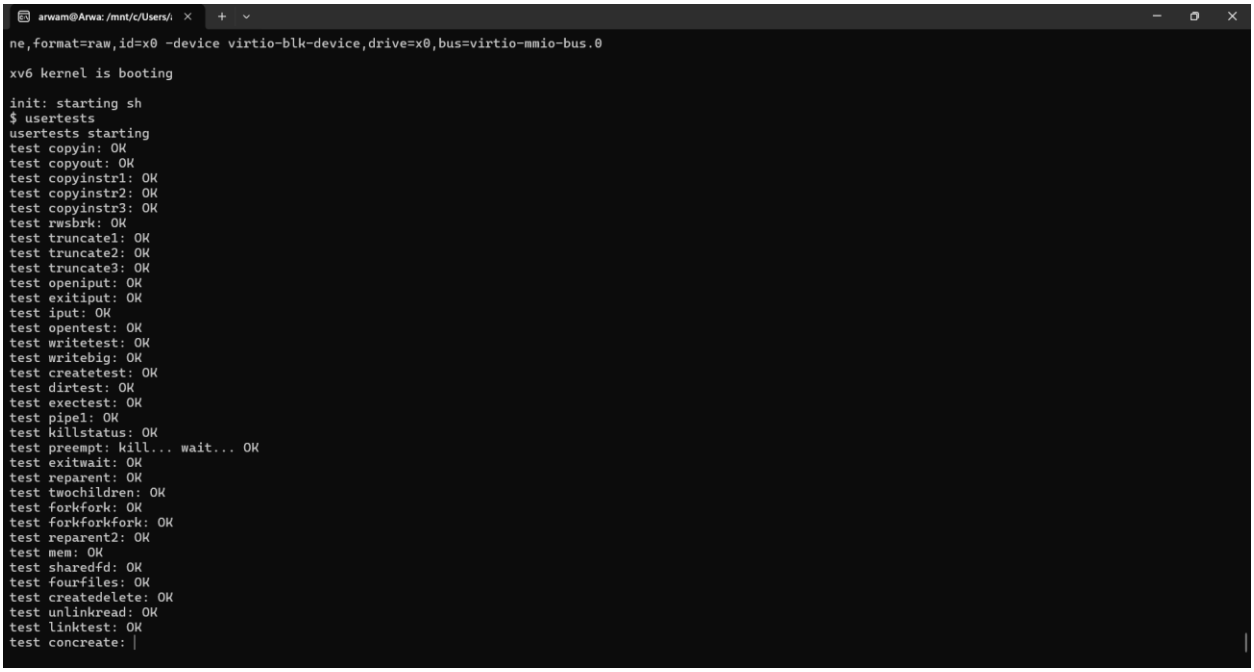

arwam@Arwa: /mnt/c/Users/
GNU nano 7.2 kernel/spinlock.c *
// Mutual exclusion spin locks.
#include "types.h"
#include "param.h"
#include "memlayout.h"
#include "spinlock.h"
#include "riscv.h"
#include "proc.h"
#include "defs.h"
void
initlock(struct spinlock *lk, char *name)
{
lk->name = name;
lk->locked = 0;
lk->cpu = 0;
}
int
tryacquire(struct spinlock *lk)
{
push_off();
if(__sync_lock_test_and_set(&lk->locked, 1) != 0){
pop_off();
return 0;
}
__sync_synchronize();
lk->cpu = mycpu();
return 1;
}
// Acquire the lock.
// Loops (spins) until the lock is acquired.
void
acquire(struct spinlock *lk)
{
push_off(); // disable interrupts to avoid deadlock.
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

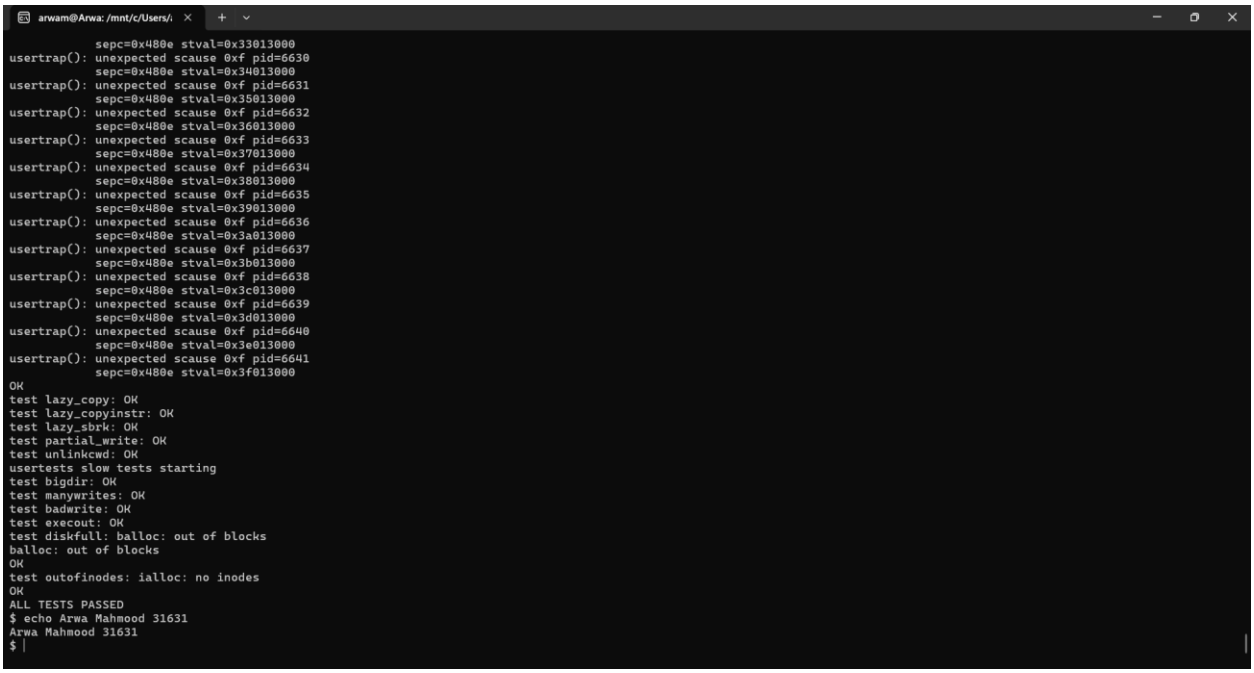

arwam@Arwa: /mnt/c/Users/
GNU nano 7.2 kernel/defs.h *
void sleep(void);
void userinit(void);
int kwait(uint64);
void wakeup(void*);
void yield(void);
int either_copyout(int user_dst, uint64 dst, void *src, uint64 len);
int either_copyin(void *dst, int user_src, uint64 src, uint64 len);
void procdump(void);
// swtch.S
void swtch(struct context*, struct context*);
// spinlock.c
void acquire(struct spinlock*);
int holding(struct spinlock*);
void initlock(struct spinlock*, char*);
void release(struct spinlock*);
void push_off(void);
void pop_off(void);
int tryacquire(struct spinlock *lk);
// sleeplock.c
void acquiresleep(struct sleeplock*);
void releasesleep(struct sleeplock*);
int holdingsleep(struct sleeplock*);
void initsleeplock(struct sleeplock*, char*);
// string.c
int memcmp(const void*, const void*, uint);
void* memmove(void*, const void*, uint);
void* memset(void*, int, uint);
char* safestrcpy(char*, const char*, int);
int strlen(const char*);
int strncmp(const char*, const char*, uint);
char* strncpy(char*, const char*, int);
// syscall.c
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

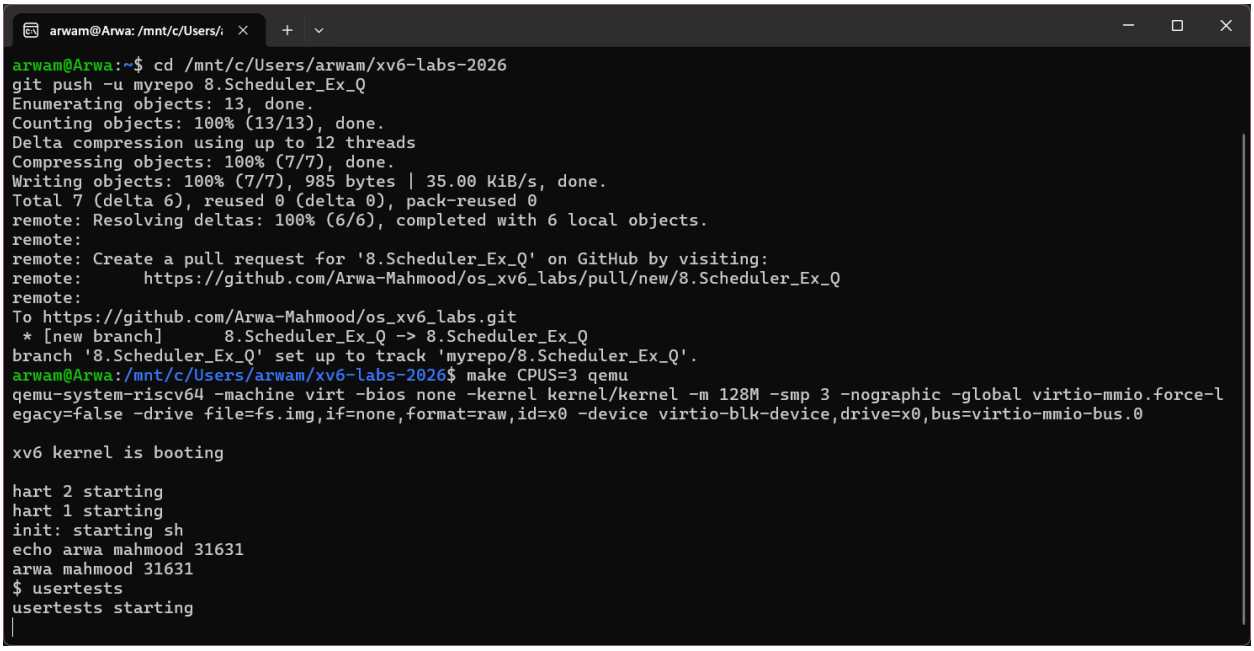

arwam@Arwa: /mnt/c/Users/
GNU nano 7.2 kernel/proc.h
// Saved registers for kernel context switches.
struct context {
uint64 ra;
uint64 sp;
// callee-saved
uint64 s0;
uint64 s1;
uint64 s2;
uint64 s3;
uint64 s4;
uint64 s5;
uint64 s6;
uint64 s7;
uint64 s8;
uint64 s9;
uint64 s10;
uint64 s11;
};
// Per-CPU state.
struct cpu {
struct proc *proc; // The process running on this cpu, or null.
struct context context; // swtch() here to enter scheduler().
int noff; // Depth of push_off() nesting.
int intena; // Were interrupts enabled before push_off()?
struct proc *prev; // thread we just switched away from; its lock must be released
};
extern struct cpu cpus[NCPU];
// per-process data for the trap handling code in trampoline.S.
// sits in a page by itself just under the trampoline page in the
// user page table. not specially mapped in the kernel page table.
// uservec in trampoline.S saves user registers in the trapframe,
// then initializes registers from the trapframe's
// kernel_sp, kernel_hartid, kernel_satp, and jumps to kernel_trap.
[Read 105 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

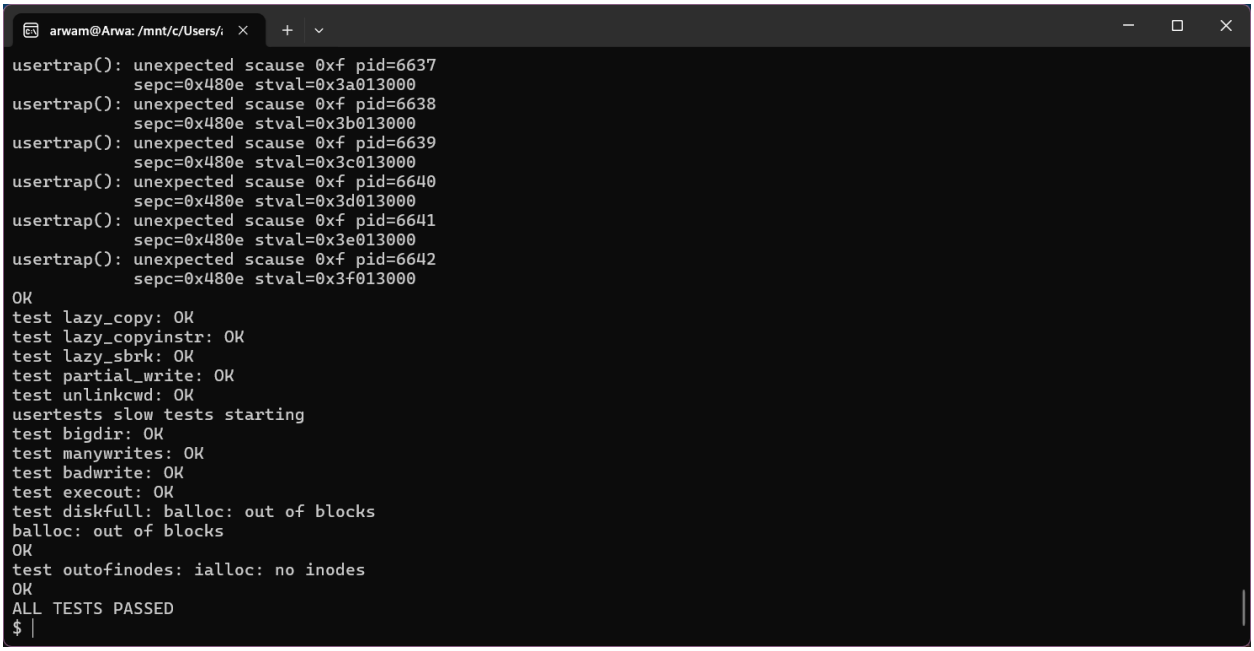

arwam@Arwa: /mnt/c/Users/
GNU nano 7.2 kernel/proc.c *
struct proc *p = myproc();
acquire(&p->lock);
p->state = RUNNABLE;
sched();
release(&p->lock);
}
void
finish_switch(void)
{
struct cpu *c = mycpu();
struct proc *pr = c->prev;
if(pr){
c->prev = 0;
release(&pr->lock);
}
}
// A fork child's very first scheduling by scheduler()
// will swtch to forkret.
void
forkret(void)
{
extern char userret[];
static int first = 1;
struct proc *p = myproc();
// Still holding p->lock from scheduler.
release(&p->lock);
finish_switch();
if (__atomic_load_n(&first, __ATOMIC_ACQUIRE)) {
// File system initialization must be run in the context of a
// regular process (e.g., because it calls sleep), and thus cannot
// be run from main().
fsinit(ROOTDEV);
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was


arwam@Arwa: /mnt/c/Users/a
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$ nano kernel/proc.c
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$ make CPUS=1 qemu   # usertests
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=. -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o kernel/proc.o kernel/proc.c
riscv64-unknown-elf-gcc -march=rv64gc -g -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=. -c -o kernel/swtch.o kernel/swtch.S
riscv64-unknown-elf-gcc -march=rv64gc -g -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=. -c -o kernel/trampoline.o kernel/trampoline.S
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=. -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o kernel/trap.o kernel/trap.c
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=. -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o kernel/syscall.o kernel/syscall.c
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=. -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o kernel/sysproc.o kernel/sysproc.c
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=. -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o kernel/bio.o kernel/bio.c
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=. -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o kernel/fs.o kernel/fs.c
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=. -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o kernel/log.o kernel/log.c
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=. -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-built


arwam@Arwa: /mnt/c/Users/a
ne,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
init: starting sh
$ usertests
usertests starting
test copyin: OK
test copyout: OK
test copyinstr1: OK
test copyinstr2: OK
test copyinstr3: OK
test rwsbrk: OK
test truncate1: OK
test truncate2: OK
test truncate3: OK
test openiput: OK
test exitiput: OK
test iput: OK
test opentest: OK
test writetest: OK
test writebig: OK
test createtest: OK
test dirtest: OK
test exectest: OK
test pipe1: OK
test killstatus: OK
test preempt: kill... wait... OK
test exitwait: OK
test reparent: OK
test twochildren: OK
test forkfork: OK
test forkforkfork: OK
test reparent2: OK
test mem: OK
test sharedfd: OK
test fourfiles: OK
test createdelete: OK
test unlinkread: OK
test linktest: OK
test concreate:


arwam@Arwa: /mnt/c/Users/
sepc=0x480e stval=0x33013000
usertrap(): unexpected scause 0xf pid=6630
sepc=0x480e stval=0x34013000
usertrap(): unexpected scause 0xf pid=6631
sepc=0x480e stval=0x35013000
usertrap(): unexpected scause 0xf pid=6632
sepc=0x480e stval=0x36013000
usertrap(): unexpected scause 0xf pid=6633
sepc=0x480e stval=0x37013000
usertrap(): unexpected scause 0xf pid=6634
sepc=0x480e stval=0x38013000
usertrap(): unexpected scause 0xf pid=6635
sepc=0x480e stval=0x39013000
usertrap(): unexpected scause 0xf pid=6636
sepc=0x480e stval=0x3a013000
usertrap(): unexpected scause 0xf pid=6637
sepc=0x480e stval=0x3b013000
usertrap(): unexpected scause 0xf pid=6638
sepc=0x480e stval=0x3c013000
usertrap(): unexpected scause 0xf pid=6639
sepc=0x480e stval=0x3d013000
usertrap(): unexpected scause 0xf pid=6640
sepc=0x480e stval=0x3e013000
usertrap(): unexpected scause 0xf pid=6641
sepc=0x480e stval=0x3f013000
OK
test lazy_copy: OK
test lazy_copyinstr: OK
test lazy_sbrk: OK
test partial_write: OK
test unlinkcwd: OK
usertests slow tests starting
test bigdir: OK
test manywrites: OK
test badwrite: OK
test execout: OK
test diskfull: balloc: out of blocks
balloc: out of blocks
OK
test outofinodes: ialloc: no inodes
OK
ALL TESTS PASSED
$ echo Arwa Mahmood 31631
Arwa Mahmood 31631
$


arwam@Arwa: /mnt/c/Users/
arwam@Arwa:~$ cd /mnt/c/Users/arwam/xv6-labs-2026
git push -u myrepo 8.Scheduler_Ex_Q
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 12 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 985 bytes | 35.00 KiB/s, done.
Total 7 (delta 6), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (6/6), completed with 6 local objects.
remote:
remote: Create a pull request for '8.Scheduler_Ex_Q' on GitHub by visiting:
remote: https://github.com/Arwa-Mahmood/os_xv6_labs/pull/new/8.Scheduler_Ex_Q
remote:
To https://github.com/Arwa-Mahmood/os_xv6_labs.git
* [new branch] 8.Scheduler_Ex_Q -> 8.Scheduler_Ex_Q
branch '8.Scheduler_Ex_Q' set up to track 'myrepo/8.Scheduler_Ex_Q'.
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$ make CPUS=3 qemu
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
echo arwa mahmood 31631
arwa mahmood 31631
$ usertests
usertests starting


arwam@Arwa: /mnt/c/Users/a
usertrap(): unexpected scause 0xf pid=6637
sepc=0x480e stval=0x3a013000
usertrap(): unexpected scause 0xf pid=6638
sepc=0x480e stval=0x3b013000
usertrap(): unexpected scause 0xf pid=6639
sepc=0x480e stval=0x3c013000
usertrap(): unexpected scause 0xf pid=6640
sepc=0x480e stval=0x3d013000
usertrap(): unexpected scause 0xf pid=6641
sepc=0x480e stval=0x3e013000
usertrap(): unexpected scause 0xf pid=6642
sepc=0x480e stval=0x3f013000
OK
test lazy_copy: OK
test lazy_copyinstr: OK
test lazy_sbrk: OK
test partial_write: OK
test unlinkcwd: OK
usertests slow tests starting
test bigdir: OK
test manywrites: OK
test badwrite: OK
test execout: OK
test diskfull: balloc: out of blocks
balloc: out of blocks
OK
test outofinodes: ialloc: no inodes
OK
ALL TESTS PASSED
$